GIANO: Guided Assistive Piano Learning Gloves

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The GIANO team has developed a prototype wearable glove system that enables accessible, low-cost piano learning without a physical keyboard, paid lessons, or dependence on visual feedback. Traditional piano education often requires expensive equipment, fees and access to lessons, and sight-based pattern recognition, which limits access for those without financial resources and low-vision and legally blind users. Music education supports motor coordination, memory, focus, and social-emotional development. GIANO offers a haptic feedback-driven, computer vision-assisted platform that delivers real-time instruction, feedback, and audio output across both free-play and guided learning modes.

At the center of the system is a Raspberry Pi 5 running computer vision algorithms that generate a virtual keyboard mask, track hand position, and manage hand positioning logic. It communicates with three Teensy 4.0 microcontrollers: two on glove-mounted PCBs and one dedicated to audio generation. Each glove integrates seven haptic motors (one per finger and two for hand direction) along with velostat pressure sensors and flex sensors that detect finger presses, movement, and articulation. By combining sensor data with camera-based tracking, the system provides finger-by-finger guidance in learning mode.

After a brief calibration, in learning mode, the Raspberry Pi issues data instruction sets containing data such as note, finger, and hand position goals. The glove-mounted Teensys activate the appropriate haptic motors through an I2C multiplexer and monitor sensors for user input. When a press is detected, they transmit the information to the Raspberry Pi, which verifies it against the expected note. All detected presses, in any mode, are also sent to the third Teensy, which produces audio using FM-based synthesis.

The prototype reliably detects single note presses, provides clear haptic cues, and generates accurate sound output. A key limitation appears during chord instruction: driving multiple motors simultaneously through the I2C multiplexer increases latency and reduces responsiveness.

Future work includes improving haptic control for multi-finger chords and creating a fully hands-free, visual-free setup process to enhance accessibility. Overall, GIANO demonstrates a portable and inclusive approach to music learning, lowering barriers for users of all abilities and backgrounds.